

Quantifying long-distance dispersal of an outbreaking insect species using trap capture data and phenology

Franck Gandiaga^{*}, Patrick M.A. James

Institute of Forestry and Conservation, John H. Daniels Faculty of Architecture, Landscape, and Design, University of Toronto, Toronto, ON, Canada

ARTICLE INFO

Keywords:

Dispersal modelling
Insect outbreaks
Phenology
Spruce budworm

ABSTRACT

Long-distance dispersal plays an important role in insect population dynamics and is a key factor in insect outbreaks. However, accurately measuring the dispersal distance of insect species can be challenging. We present a simple and reliable method to quantify dispersal capacity using insect phenology and time-series of field-capture data. This open-source method is developed in R and tested using data collected on the eastern spruce budworm (*Choristoneura fumiferana*). Empirical data were collected in the summer of 2014 in Québec, in forest areas defoliated during the ongoing outbreak. Sampling dates were informed by BioSIM phenological models and government of Canada weather data where adult moths were sampled before, during, and after expected flying periods. A sensitivity analysis showed that the uncertainty of these models due to weather station interpolation did not affect general trends. Individuals sampled outside of their expected phenological window were assumed to be migrants. Pairwise Euclidean distances were measured between the sampling locations of these migrants and all potential areas of origin as predicted by BioSIM at the time of their sampling. The shortest distances were then selected to generate dispersal kernels. According to the dispersal kernels derived using this method, most flying individuals captured outside of their phenological window travelled up to 450 km, matching previous work on eastern spruce budworm dispersal. Here, we were able to demonstrate that with known phenology of a species and relatively simple to collect data (presence of flight capable individuals), it is possible to characterize insect dispersal using a relatively simple algorithm and spatial model. This method can be easily applied to other outbreaking species and used to inform management decisions in a variety of systems where outbreaks represent a risk to resources of interest.

1. Introduction

Outbreaks of forest insect pests are a recurring natural disturbance in northern forests (Baskerville 1975, Royama 1984, Myers 1993, MacLean 2019) and are an integral component of forest ecosystem functioning (Risser 1987, Volney and Fleming 2000). Many forest pest species exhibit outbreaks – that is, temporally periodic and spatially synchronous fluctuations in population density over large geographical areas (Peltonen et al. 2002). While many outbreaking species are native to Canada's northern forests, outbreaks can have negative effects on local ecosystems and economies (Sturtevant et al. 2015). A clear understanding of how outbreaks develop in space and time is needed to anticipate, forecast, and manage these negative effects (Pureswaran et al. 2016).

The eastern spruce budworm (SBW) is an outbreaking native Lepidopteran that defoliates balsam fir (*Abies balsamea*, (L.) Miller) and spruce species (*Picea spp.*) stands in the boreal and sub-boreal forests of

North America (Royama 1984, Sturtevant et al. 2013). SBW outbreaks impact forest ecosystem processes (Dymond et al. 2010, Boulanger et al. 2017, James et al. 2017) as well as the forest industry and forest associated communities (Chang et al. 2012, MacLean 2016, Boulanger et al. 2017). The current SBW outbreak began in 2006 and has affected more than eight million hectares of forest in Québec, Ontario, and New Brunswick (Ministère des Forêts, 2015; Ministère des Forêts, 2020). In response to this outbreak, a novel early intervention program was developed with the aim of mitigating the negative effects of defoliation (Johns et al. 2019, MacLean et al. 2019).

Effective deployment of early intervention strategies is sensitive to our knowledge and understanding of the spatial and temporal dynamics of SBW outbreaks (Régnière et al. 2001, Johns et al. 2019). Among the most important processes determining these dynamics is dispersal of adult moths, although it is also one of the least well understood (Pureswaran et al. 2016, Johns et al. 2019). Dispersal influences spatial synchrony among separated outbreaking populations (Anderson and

^{*} Corresponding author.

<https://doi.org/10.1016/j.foreco.2023.121187>

Received 3 March 2023; Received in revised form 5 June 2023; Accepted 7 June 2023

Available online 22 June 2023

0378-1127/© 2023 Elsevier B.V. All rights reserved.

Sturtevant 2011, Larroque et al. 2019) and can shift populations from endemic to epidemic states resulting in large scale outbreaks (Boulanger et al. 2017, Régnière and Nealis 2019, Larroque et al. 2020).

Dispersal is difficult to quantify (Ims and Andreassen 2005), especially when studying small-bodied organisms such as insects (Anderson and Sturtevant 2011, Osborne and Loxdale 2014). Both direct and indirect methods have been used to quantify SBW adult dispersal. One of the earliest direct approaches involved aerial trapping of migrating moths using tethered balloons or towed nets attached to aircraft (Greenbank et al. 1980). Dispersal parameters derived from this more than 40 year-old study (Greenbank et al. 1980) remain essential for atmospheric modelling of SBW dispersal (Sturtevant et al. 2013). More recently, surveillance radar and high-resolution three-dimensional weather data have been used to directly track clouds of moths during mass flight events (Boulanger et al. 2017). Combined atmospheric modelling and weather data have also been used to characterize SBW dispersal during such flight events (Garcia et al. 2022). Although direct methods are useful to characterize dispersal, they are challenging to implement because of the costs associated with such large-scale sampling (Greenbank et al. 1980) and often require protocols tailored to a specific species and restricted to specific and short-term local events (Boulanger et al. 2017, Garcia et al. 2022).

Indirect methods for estimating SBW dispersal include analysis of egg-moth ratios (Nealis and Régnière 2004), comparisons between trap captures and SBW phenological models (Sturtevant et al. 2013), variation in sex ratio and morphology (Rhainds 2015), as well as light-traps (Rhainds et al. 2022). Indirect methods are useful to demonstrate that immigration has occurred but do not necessarily identify the source population from which immigrants dispersed, nor do they contribute to our understanding of the distances travelled.

Monitoring of spruce budworm populations across Canada and the eastern US relies on pheromone baited moth traps (Carleton et al. 2020). These traps are typically installed in the late spring around the time of initial moth eclosion. Traps are then collected at the end of the flight season and male moth counts are used to estimate local population densities. While cost effective and efficient when considering regional and landscape scale monitoring objectives, such traps may not accurately reflect local population dynamics due to the high dispersal capacity of the SBW (Sturtevant et al. 2013). As such, it is important to continue developing new approaches to characterise SBW dispersal.

Repeated collection from moth traps can help overcome this challenge of distinguishing regional from local population dynamics. Time series of male moth captures throughout the flight season can be used to separate local from migrant moths based on phenology. Once a captured moth is classified as either a resident or a putative migrant (i.e., a moth captured out of local phenological synchrony) one can assign potential migrants to a location of origin. By repeating this procedure with many putative migrant moths from many locations, one can develop a distribution of likely dispersal distances which can be used to refine our understanding of spatial outbreak dynamics to inform early intervention strategies (Johns et al. 2019, MacLean et al. 2019). For example, a greater understanding of the probability of dispersal at given distances can improve the identification and classification of potential and actual hotspots and guide the early deployment of mitigation treatments such as insecticide application (e.g., *Bacillus thuringiensis* var. *Kurstaki* (Btk); Fuentealba et al. 2019).

In this paper, we develop and present a new method to characterise patterns of dispersal in outbreaking forest insect pests using time series of moth capture data and phenology, using the spruce budworm as a motivating example. Our method combines time series of male moth collection within a single season with a spatial phenological model that classifies individuals as either residents or potential migrants. We then assign potential migrants to likely source locations based on phenology and landscape geometry, while subject to a range of spatial constraints. Repeating this process for all identified potential migrants permits the generation of a distribution of probable distances travelled (i.e., a

dispersal kernel) that can be used in other downstream modelling and forecasting applications. We also report on the model sensitivity to different assumptions related to the range of likely geographic source populations, host species, and our confidence in the spatial and temporal accuracy of the phenological model and collection data. Although we develop and present this work in the context of the spruce budworm, our approach is applicable to other mobile insect pests in both forest and agricultural systems.

2. Methods

2.1. Model concept and design

The method we present quantifies the dispersal capacity of a mobile insect pest species in the form of a dispersal kernel. The main premise of our approach is that we can identify potential migrant moths and assign them to a likely source of origin based on phenology (i.e., expected emergence based on local environmental conditions; Fig. 1). We can then measure the geographic distance between the location at which a potential emigrant was captured and its most likely phenologically-determined source location. Repeating this procedure for numerous migrants generates a frequency distribution of likely dispersal distances which can be fit to a statistical distribution which we refer to as a dispersal kernel. The raw data on which this procedure operates are geographically referenced time series of adult moth captures and a daily spatial phenological model of predicted moth emergence.

There are five discrete steps to this process (Fig. 2). The first step consists of collecting moths through time at multiple locations within a season. Each date/location combination is considered a *sampling event*. These data may vary in terms of the temporal frequency with which they are collected, although in general, greater frequencies of collection are to be preferred.

The second step involves generating predictions of the expected timing of moth emergence across the full extent of interest using the phenological model BioSIM (Régnière and St-Amant 2007, Régnière et al. 2012). Our application of BioSIM involves calculating the probability of moth emergence for every day in the potential flight season across the spatial extent of our study area at a spatial resolution of 10×10 km (Fig. 4) using environment Canada weather data. Modelled confidence intervals (CIs) of emergence probability (e.g., 95% CI) are then used to classify collected individuals as either residents or migrants. If a moth was collected outside of this local phenological interval (i.e., phenological asynchrony), it is classified as a migrant. If collected within those bounds (i.e., phenological synchrony), the moth is classified as a

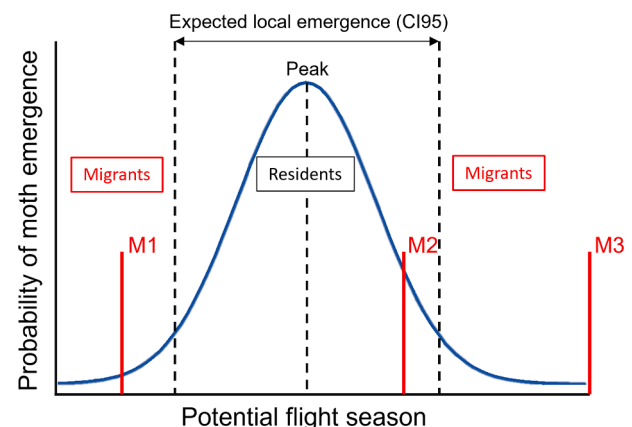


Fig. 1. Curve representing for each sampling location the probability of moth emergence for every day through the potential flight season using BioSIM. The red markers represent the approximate times of sample collection for each site. The markers indicate sampling before (M1), during (M2) and after (M3) confirmed presence.

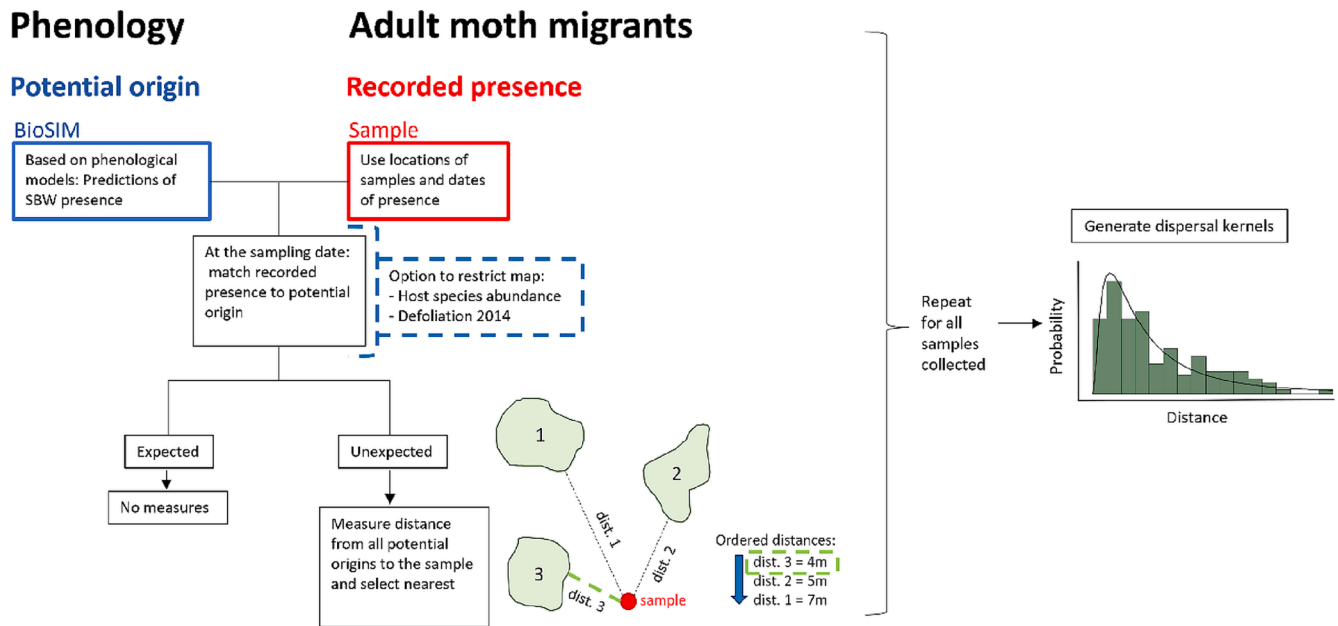


Fig. 2. Conceptual representation of the workflow developed to generate dispersal kernels using our proposed method. Step 1: Sample individuals in the area throughout their flight season. Step 2: Generate predictions of the expected timing of moth emergence across the full extent of interest using the phenological model BioSIM. Step 3: Match potential origins with recorded presence at the date of the sampling. Step 4: If the date of sampling matches an expected presence, then the method stops. If the presence of individuals is unexpected (phenologically asynchronous), we measure all distances from potential origins to the sample location. All distances are compared and the nearest is selected. This step is repeated for all the samples collected throughout the season. Step 5: All distances selected at the end of Step 4 are used to generate dispersal kernels.

resident (Fig. 1).

In step three, BioSIM-derived raster files of daily emergence probabilities are used to evaluate potential source locations of migrant moths. Potential moth source locations are those within the study area that are in phenological synchrony with the location where the migrant moth was captured. In the simplest application of this step, all locations in phenological synchrony at the time that migrant moths were captured are considered as potential source locations.

In step four, we calculate Euclidean distances between the locations at which migrants were identified and the nearest point in phenological synchrony. Here, we assume that the nearest phenologically synchronous location is the most likely dispersal source. Repeating this analysis for all phenologically asynchronous moths (both early and late) we generate a distribution of potential dispersal distances based on this assumption geographic proximity. While the simplest assumption is that potential source locations are defined solely by phenology, it is also possible to restrict the set of potential host locations based on host availability or areas of known outbreak, habitat suitability, or other environmental factors.

Finally, in the fifth step, the distribution of potential dispersal distances generated in step four is used to fit a dispersal kernel (Fig. 3). Here, we fit the data to a probability density function that best describes the distribution of dispersal distances of a given set of individuals (Nathan et al. 2012). In this study, we compared four distribution models: Weibull, log-normal, exponential, and Gaussian. Dispersal kernels were fit using Ordinary Least Squares regression (Hirsch et al. 2012; Appendix S1). We determined which distribution best fit our data using the Akaike information criterion (AIC; Akaike 1974).

2.2. Model implementation

We developed and tested this method using the eastern spruce budworm. Adult male moths were collected in Québec at three different times in the summers of 2012, 2013, and 2014 (Fig. 4) using Multipher1 traps (Jobin and Coulomb 1988) baited with a pheromone lure (Distributions Solida) and containing a vapour tape insecticide (Dichlorvos

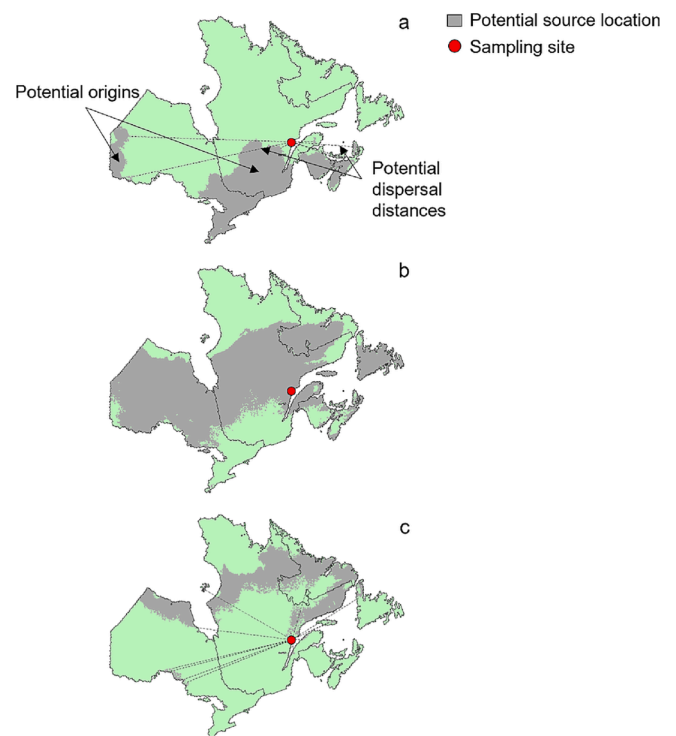


Fig. 3. Visual representation of the output generated by our method on a map of Ontario, Québec, New Brunswick, Nova Scotia, Newfoundland, and Labrador. Each map represents a) a M1 asynchronous sample, b) a M2 synchronous sample and c) a M3 asynchronous sample. The grey areas represent the potential presence of adults as predicted by BioSIM while the red dot represents the actual location at which individuals were collected. The dotted lines represent the shortest distance between each potential origin patch and the sampling site.

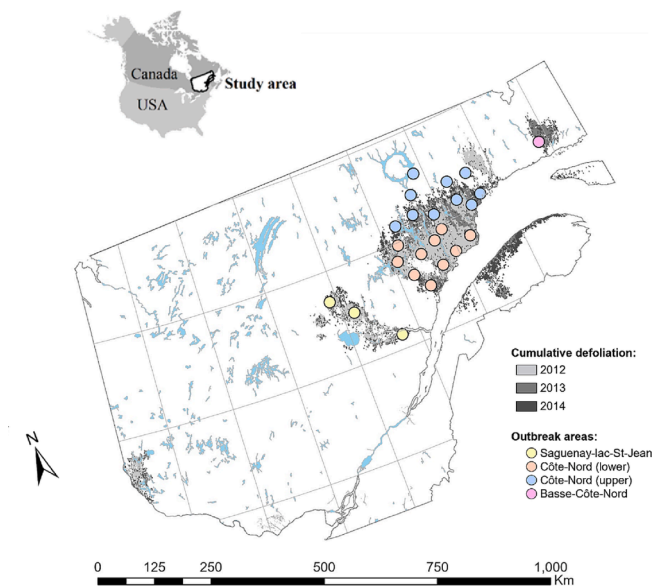


Fig. 4. Study area showing the locations of the 24 sampling sites within the province of Québec (Canada) with the extent of spruce budworm defoliation as of 2012 (light gray), 2013 (medium gray) and 2014 (dark gray). The colors indicate the different outbreaking areas. At each site we sampled moths three times during the season 1) before it was likely that moths were flying locally; 2) at the peak flying time; and 3) after moths should have stopped flying locally.

10%; Hercon). Traps were deployed at approximately 2 m above ground in host forests across 24 sites in known outbreak areas affected by defoliation (Fig. 4). Timing of moth trap deployment and collection was informed using the BioSIM phenological model (Régnière et al. 2012) executed using raster grid cells at a spatial resolution of 10 km × 10 km.

Pheromone-baited moth traps were initially deployed in early June, ~2 weeks before expected moth flight period. Traps were then emptied at three times. We targeted putative early migrant moths captured prior to the predicted phenological window (M1) by emptying the traps 1–2 weeks prior to the phenologically predicted period (95% confidence interval) of moth emergence. Traps were emptied a second time just after (<1 week) the period of predicted moth emergence to sample putative resident moths (M2). Finally, moths (if present) were collected from traps a third time (M3) approximately two weeks after M2 collection. Due to logistical constraints including significant distances between sites, shifting phenology, and unexpected site disturbances (e.g., forest fires) it was not possible to visit all sites for all three sampling periods. Pheromone trapped moths from the Armagh site were generously provided by Dr. Jacques Régnière (Canadian Forest Service, NRCan).

Final classification of captured moths as M1, M2, or M3 was determined using final phenology based on observed (vs. predicted) weather (Fig. 2). Using BioSIM, we calculated the 95% confidence interval of expected emergence timing for each sample site. Moths that were presumed to have entered the trap before the 2.5th percentile of predicted local moth emergence were classified as early potential migrants (M1). Moths that entered the trap during the period of the 95% confidence interval were classified as residents that had emerged from local larvae (M2). Finally, moths that entered the trap after the 97.5 percentile of predicted local moth emergence were deemed to be potential late migrants (M3) and their likely dispersal distances were calculated accordingly.

2.3. Model sensitivity

We assessed the sensitivity of our approach to different degrees of precision in the phenological classification, use of different host species

to drive the phenological model, as well as to different spatial constraints that determine potential source locations. With regards to phenological classification of moths as migrants or residents, we assessed how changing the phenological confidence interval over which individuals were classified as residents from 95% to 99% (~2–3 days later). Extending the confidence interval moved some individuals from the M1 to the M2 class as well as some from the M3 to the M2 class; more individuals are classified as residents with a wider confidence interval.

We also compared phenological predictions based on each of the two host species (i.e., balsam fir vs. white spruce). The BioSIM models developed for each host species vary according to each host's phenological timing of budburst, bud abundance, and bud weight that directly impact spruce budworm population density (Régnière and You 1991).

Finally, we examined the sensitivity of our approach to three levels of spatial constraints on potential source locations. First, the no spatial constraint model only included areas in phenological synchrony as potential sources (Fig. 3). Second, we constrained potential source locations to those that were in phenological synchrony and contained at least 1 % of host (i.e., balsam fir or white spruce) based on remotely sensed data (Beaudoin et al. 2014). Our third level of constraint restricted potential source locations to only those cells that were in phenological synchrony and had been identified as defoliated in the year of sampling (Ministère des Forêts, 2015; Fig. 4).

3. Results

Across the three years, a total of 151 sampling events during which time traps were visited and emptied took place (Table 1). Of these sampling events, 38 samples contained moths before a period of predicted moth emergence, 28 samples contained moths at the expected time of emergence, and 85 contained moths after the expected period of emergence. As this study is focused on assigning phenologically asynchronous moths to potential source locations, only the 114 sampling events that captured phenologically asynchronous moths were analysed. Out of a total of 115,106 moths collected, 73,689 we identified potential migrant moths, of which 7,499 were identified as early, and 66,200 were identified as late (Table 1).

Dispersal kernels were generated using the sampling locations and the phenological models of balsam fir and white spruce, the two main host species of the SBW. Differences in phenology of these two species influences how collected moths are classified as either residents or migrants. Therefore, we sought to compare dispersal estimates based on these different models of expected timing of moth emergence. The range, median, and mean dispersal distances estimated for each host species and different spatial constraints can be found in Table 2. The removal of individuals moths that were somewhat uncertain in terms of their assignment to a migrant vs. resident class using the 99% confidence interval did not affect our trends (not shown). The best fit for the dispersal kernels was a log-normal distribution for the phenological synchrony (no spatial restriction) and host species models (Fig. 5a – d) and a Weibull distribution for the defoliation as of 2014 models (Fig. 5e and f). Dispersal distances were slightly longer using spruce phenology but overall, both models (i.e., balsam fir and spruce) found similar results (Table 2).

The distribution of dispersal distances was similar between the no spatial constraint model (phenological synchrony only) and the host species spatial constraint. In both scenarios, dispersal estimates ranged from 1 to 706 km and 14 to 706 km, respectively. Likewise, mean and standard deviation in dispersal distances were 273±41 and 273±42 respectively. In these two types of spatial constraint, most dispersal events ranged in between 20 and 450 km (~65%; Table 2), a third of them were above 450 km, and only a few were under 20 km (Table 2).

In contrast, we noted a marked influence of constraining source locations to defoliated areas. Restricting potential source locations to defoliated areas did not decrease the maximum estimated dispersal distance but did decrease the average distance from 275 km to 130 km

Table 1

Locations of each site sampled from 2012 to 2014 and the number of individuals collected for each year of sampling for the 95% phenological confidence. Multiple values per trap indicate multiple sampling dates at that location. Categories M1 (early) and M3 (late) indicate putative migrant moths captured prior to or after the predicted phenological window, respectively while M2 indicates putative resident moths captured within the expected phenological window.

Sites	Outbreak Area	Lat.	Long.	CI95								
				2012			2013			2014		
				M1	M2	M3	M1	M2	M3	M1	M2	M3
1	Saguenay-Lac-St-Jean	48.28	−70.54	.	.	.	.	.	754	.	.	1188
				.	.	.	.	.	11	.	.	.
2		49.03	−71.64	.	25	2324	.	.	2746	.	61	852
3	Côte-Nord lower	49.40	−72.23	.	.	.	.	.	5	.	.	.
				.	.	.	.	.	1215	.	31	1834
4		48.96	−69.19	5	.	1132	.	1070	15	18	3176	11
5		49.24	−68.60	2	.	2220	.	2687	7	288	.	2089
				.	.	.	.	.	.	.	.	7
6		49.28	−69.53	86	.	1338	.	1341	19	14	.	1780
7		49.39	−68.07	1	.	572	15	540	2	9	410	1
				.	.	.	.	.	.	.	.	10
8		49.55	−67.48	.	489	3	89	2511	12	712	.	751
9		49.61	−69.08	15	.	2805	.	4827	.	.	217	4643
				.	.	2	.	.	.	.	.	6
10		49.65	−69.84	.	2	2140	.	.	3740	6	.	26
11		49.75	−68.54	15	.	913	158	2738	2	60	1060	8
				.	.	.	.	.	.	.	.	.
12		49.90	−68.20	.	.	1098	.	3155	24	1023	.	2650
13	Côte-Nord upper	49.95	−69.64	5	.	2	.	5277	252	4	.	3387
				.	.	.	.	.	.	.	.	40
14		50.09	−67.05	.	1028	20	248	.	3525	247	.	1992
15		50.22	−66.66	.	526	.	215	.	1126	398	.	1
				.	.	.	.	.	6	.	.	.
16		50.23	−68.24	.	374	.	.	3254	190	48	2690	.
17		50.31	−67.40	3	2291	8	418	.	4256	1472	.	2
				.	.	.	.	.	17	.	.	.
18		50.32	−69.49	.	.	.	.	.	.	3	1123	96
19		50.39	−68.85	2	.	104	35	.	4018	.	90	2009
				.	.	5	.	.	78	.	.	1
20		50.72	−67.45	.	216	3	.	.	.	184	.	1803
21		50.72	−66.80	2	198	1	220	.	348	154	.	151
				.	.	.	.	.	1	.	.	2
22		50.77	−68.66	.	.	248	24	.	1743	39	.	.
23	Basse-Côte-Nord	51.14	−68.31	.	.	1	.	.	8	.	.	.
				.	.	263	.	.	.	64	.	561
24		50.62	−64.30	.	.	2	.	.	.	.	.	.
				.	.	.	.	.	413	1198	.	.

Table 2

Range, median and mean of the distances of dispersal events (in km) as measured using our method and distribution of the dispersal events (in %) per category of distances (<20 km, 20–450 km, greater than 450 km). We tested the phenological models of Balsam fir and Spruce species. We tested three levels of spatial constraints, 1) no spatial restriction (or phenological synchrony of SBW), 2) host species distribution and 3) defoliated areas as of 2014. We applied a 95% confidence interval to calculate all the probabilities of moth occurrence used to measure these distances.

Phenological model	Spatial constraint	Estimated dispersal distance (km)			Dispersal events (%)		
		Range	Median	Mean ± CI95%	< 20 km	20–450 km	> 450 km
Balsam fir	No spatial constraint	1–706	161.38	273±42	4	65	30
	Host species	14–706	176.19	273±41	4	65	31
	Defoliated area	14–695	100.66	131±24	5	94	1
White spruce	No spatial constraint	1–809	159.25	291±45	3	62	35
	Host species	12–821	164.97	289±43	3	63	34
	Defoliated area	14–695	93.64	121±24	5	93	1

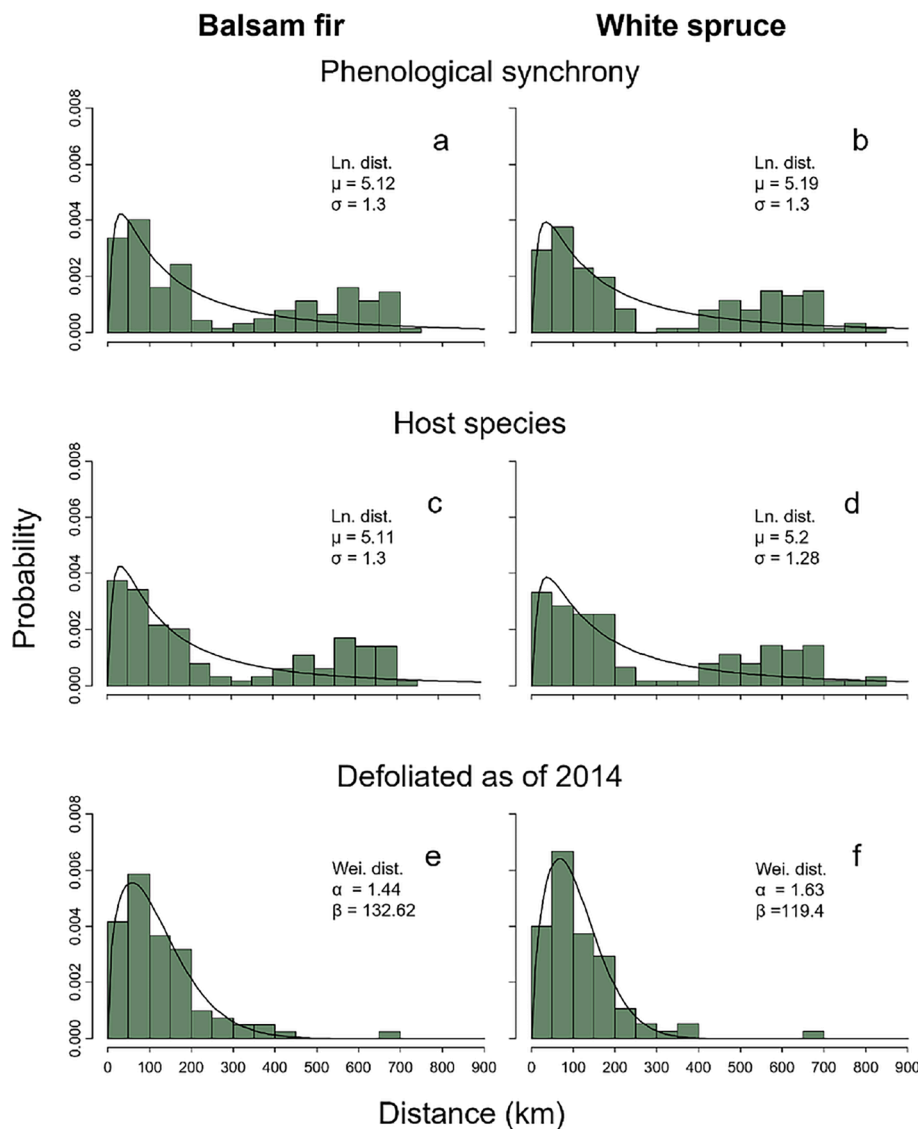


Fig. 5. Histograms of estimated distance to source for potential migrant moths with a fitted distribution curve. We used Akaike information criterion to determine which function fitted the observed data best (Ln = Lognormal, μ = location, σ = shape, Wei = Weibull, α = shape, β = scale). The graphs represent the probability of distance traveled calculated using both Balsam Fir (a,c,e) and White Spruce (b,d,f). Each level represent a different landscape hierarchy from the maximum area of no spatial constraint (or phenological synchrony; a,b) to the host species distribution area (c,d) to the known area of cumulated defoliated area as of 2014 in Québec (e,f). We applied a 95% confidence interval to calculate all the probabilities of moth occurrence used to produce these histograms.

and increased the proportion of dispersal events in the 20–450 km range from 65% to 95%. Likewise, we found a major decrease in the frequency of long-distance dispersal events greater than 450 km (i.e., from 35% to 1% or less; Table 2). While the maximum observed distances traveled in the less restrictive spatial constraints went beyond what was previously recorded, the average distance traveled as well as the proportion of distances most traveled across all levels of spatial constraints was within the 20 to 450 km range previously reported in this system (Greenbank et al. 1980, Boulanger et al. 2017).

4. Discussion

Using a combination of moth capture data and phenological prediction, we demonstrate a method to derive dispersal kernels for the spruce budworm. The method we developed is both general and flexible and can be applied to examine dispersal at a scale comparable to previous extensive empirical studies (Greenbank et al. 1980). We also demonstrate the sensitivity of the method to different assumptions regarding potential source locations and observed some differences in estimated dispersal in response to these different assumptions.

The parameterized dispersal kernels correspond well with previous work in this system (Greenbank et al. 1980, Boulanger et al. 2017, Garcia et al. 2022). Individuals moths have been observed dispersing

200 km (Boulanger et al. 2017) and up to 450 km (Greenbank et al. 1980). Similarly, previous simulation work demonstrated that moth dispersal can reach a maximum of 370 km, with a peak in male SBW dispersal probability close to 100 to 240 km depending on the event considered (Garcia et al. 2022). Our findings that most individuals dispersed between 0 and 450 km indicate that our approach using trap captures and phenology can produce similar estimates of SBW dispersal. However, in using pheromone traps, we restrict our inference regarding spruce budworm dispersal to males only, which may not be applicable to female dispersal dynamics (Rhainds et al. 2019).

Dispersal estimates based on phenological models of balsam fir and spruce species did not differ significantly (Fig. 5). We noted slightly longer maximal dispersal distances when using the spruce phenology for resident vs. migrant classification (Table 2), but similar average distances traveled (Table 2). These observations could be explained by slight variations in host species phenology. Balsam fir budburst is closely synchronized with peak emergence of overwintering SBW and tends to happen earlier than white spruce by a few days making balsam fir attractive slightly before white spruce (Greenbank 1963). As such, the spruce species phenological model predicts that a given area will be favorable for adult moth emergence later compared to the balsam fir model which leads to longer dispersal distances.

4.1. Model sensitivity and study limitations

In this initial expository treatment of our method, we made several simplifying assumptions including the use of Euclidean distances between potential sources and moths sampling locations. Previous studies have demonstrated the impact of spatial environmental factors on SBW dispersal such as wind (Greenbank et al. 1980, Anderson and Sturtevant 2011) temperature (Sanders et al. 1978, Régnière and You 1991, Sturtevant et al. 2013), and precipitation (Larroque et al. 2019). Future work will aim to better incorporate environmental variation through the use of effective distances (i.e., least-cost paths) between locations (Tischendorf and Fahrig 2000). An additional possibility would be to incorporate directional wind data from the NOAA HYSPLIT model (Stein et al. 2015, Garcia et al. 2022) to model the influence of air currents and trajectories on dispersal between potential origins and sampling locations (Garcia et al. 2022).

Another source of uncertainty in our findings stems from our reliance on an interpolated weather model to derive our phenological timings. BioSIM is an excellent model, although there are known challenges associated with its predictions when interpolating at distances that are far from weather stations (Dirks et al. 1998, Mair and Fares 2011, Ly et al. 2013). Future work will include examining the accuracy and spatial scale of uncertainty in these phenological predictions in remote areas such as the boreal forest, assessing the sensitivity of our assignment model to such variation, and investigating alternate methods for weather interpolation.

Finally, our inferences regarding dispersal may be sensitive to the relative timing and frequency of data collection. Data were collected at a relatively coarse temporal frequency; three times per location through the flight season, albeit over a very large spatial extent. Traps were typically emptied approximately two weeks after installation and as a result, the date at which the individuals were collected does not necessarily reflect the date at which the adult moth entered the trap. In this procedure, we assume that all individuals collected in a trap dispersed at the same time as we have no other information on relative timing of arrival. The delay between arrival and sampling of the individuals could lead us to either underestimate the dispersal distance early arriving individuals (M1) or overestimate the dispersal distance of late migrants (M3). We hope to refine our estimates of trap arrival timing in future work.

Finally, the coarse temporal resolution of sampling makes it difficult to identify dispersal events that occur between locations that are in phenological synchrony. For example, our method does not identify the mass exodus flight described in Garcia et al (2022), as the source and destination locations from this event were in phenological synchrony. Future work will involve using data collected at a finer temporal frequency (i.e., Budworm Tracker citizen science data; Carleton et al. 2020) that will allow us to better distinguish residents from migrants, and improve the precision and accuracy of the dispersal estimates presented here. Despite this potential source of bias, the omission of acute mass exodus flights between phenologically similar regions had little influence on our dispersal kernel estimates. The trends we identified (Table 2) are very similar to values reported in other studies of SBW moth migration (Greenbank 1984, Garcia et al. 2022, Boulanger et al. 2017).

4.2. Model flexibility and future applications

Our approach can be deployed to characterize dispersal over range of spatial extents and spatial resolutions. The demonstration of our approach adopted a provincial scale, in agreement with the expected dispersal capacity of the organism in question (SBW) and the spatial extent of its effects. Likewise, the method is flexible with regards to the spatial resolution, or grain, of spatial information used to model and constrain possible source locations and dispersal pathways.

While focused on the spruce budworm, this method can be applied to

other outbreaking pest systems. Many of the processes determining SBW phenology and dispersal, on which our method is based, are common to other pest species such as the western spruce budworm (*Choristoneura freemani*, Razowski), jack pine budworm (*Choristoneura pinus*, Freeman), mountain pine beetle *Dendroctonus ponderosae*, Hopkins) or the emerald ash borer (*Agrilus planipennis*, Fairmaire). Outbreaking species behave similarly during epidemic cycles (Peltonen et al. 2002, MacLean 2016). Individuals emerging in high-density areas with low levels of resources available will migrate to the nearest available patch (Greenbank 1963, Rhainds et al. 2002, Mazzi and Dorn 2012). For most insect species, their phenological cycle and emergence are mostly driven by environmental factors with degree days being a good indicator (Herms 2004, Crimmins et al. 2020).

The method presented here is based on phenological models implemented in the freely-available software BioSIM as well as data on the degree days necessary for moth emergence (Régnière et al. 2012). Such data is already available in BioSIM for several outbreaking species of interest including the spruce budworm, spongy moth (Régnière et al. 2012), emerald ash borer, and the Asian longhorn beetle (*Anoplophora glabripennis*; for a more exhaustive list see Crimmins et al. 2020). By combining phenological models with empirical, field-collected time-series of the presence/absence of flight capable individuals, this tool is both powerful and adaptable to quantify dispersal in a range of taxa. Its versatility makes it applicable to other systems where phenology is known (or can be known) and where time series of adult trap capture can be obtained. Our method can be easily adapted to other types of systems such as agricultural or urban landscapes.

5. Conclusion

The method we developed in this study can reliably provide us with greater understanding of the probability of dispersal at given distances of an outbreaking species. This tool can in turn be integrated into early intervention programs (e.g., EIS; MacLean et al. 2019) to complement already used approaches such as improving the efficiency of L2 larvae monitoring in the case of SBW outbreaks (Johns et al. 2019) to improve the detection of hotspots and the earlier deployment of adapted treatments.

We were also able to demonstrate that with known phenology of a species and relatively simple to collect empirical data (presence of flight capable individuals), it is possible to characterize insect dispersal using a simple algorithm and spatial model. Furthermore, the method was designed to incorporate complementary data such as advanced weather models or spatial habitat suitability data while being also flexible in terms of spatiotemporal resolution, allowing users to study a wide range of dispersal events from short-term local events to long-term seasonal dispersal across landscapes. Due to its versatility and scalability this method could also be easily applied to other outbreaking species and used to inform management decisions in a variety of systems where outbreaks represent a risk to resources of interest.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was supported by an NSERC Discovery Grant to PMAJ, and the NRCan supported Early Intervention for Spruce Budworm Outbreaks. We thank Sophie Wilkinson for feedback on an earlier version of

the manuscript.

References

- Akaike, H., 1974. A new look at the statistical model identification. *IEEE transactions on automatic control* 19 (6), 716–723.
- Anderson, D.P., Sturtevant, B.R., 2011. Pattern analysis of eastern spruce budworm *Choristoneura fumiferana* dispersal. *Ecography* 34 (3), 488–497.
- Baskerville, G.L., 1975. Spruce Budworm: Super Silviculturist. *The Forestry Chronicle* 51 (4), 138–140.
- Beaudoin, A., Bernier, P.Y., Guindon, L., Villemaire, P., Guo, X.J., Stinson, G., Bergeron, T., Magnussen, S., Hall, R.J., 2014. Mapping attributes of Canada's forests at moderate resolution through kNN and MODIS imagery. *Canadian Journal of Forest Research* 44 (5), 521–532.
- Boulanger, Y., Fabry, F., Kilambi, A., Pureswaran, D.S., Sturtevant, B.R., Saint-Amant, R., 2017. The use of weather surveillance radar and high-resolution three dimensional weather data to monitor a spruce budworm mass exodus flight. *Agricultural and Forest Meteorology* 234–235, 127–135.
- Carleton, R.D., Owens, E., Blaquière, H., Bourassa, S., Bowden, J.J., Candau, J.-N., DeMerchant, I., Edwards, S., Heustis, A., James, P.M.A., Kanoti, A.M., MacQuarrie, C.J.K., Martel, V., Moise, E.R.D., Pureswaran, D.S., Shanks, E., Johns, R. C., Kerr, J., 2020. Tracking insect outbreaks: a case study of community-assisted moth monitoring using sex pheromone traps. *Facets* 5 (1), 91–104.
- Chang, W.-Y., Lantz, V.A., Hennigar, C.R., MacLean, D.A., 2012. Economic impacts of forest pests: A case study of spruce budworm outbreaks and control in New Brunswick, Canada. *Canadian Journal of Forest Research* 42 (3), 490–505.
- Crimmins, T. M., K. L. Gerst, D. G. Huerta, R. L. Marsh, E. E. Posthumus, A. H. Rosemartin, J. Switzer, J. F. Weltzin, L. Coop, N. Dietschler, D. A. Herms, S. Limbu, R. T. Trotter, M. Whitmore, and K. Macaluso. 2020. Short-Term Forecasts of Insect Phenology Inform Pest Management. *Annals of the Entomological Society of America* 113:139-148.
- Dirks, K.N., Hay, J.E., Stow, C.D., Harris, D., 1998. High-resolution studies of rainfall on Norfolk Island: Part II: Interpolation of rainfall data. *Journal of Hydrology* 208 (3-4), 187–193.
- Dymond, C.C., Neilson, E.T., Stinson, G., Porter, K., MacLean, D.A., Gray, D.R., Campagna, M., Kurz, W.A., 2010. Future spruce budworm outbreak may create a carbon source in Eastern Canadian forests. *Ecosystems* 13 (6), 917–931.
- Fuentealba, A., Dupont, A., Hébert, C., Berthiaume, R., Quezada-García, R., Baucé, É., 2019. Comparing the efficacy of various aerial spraying scenarios using *Bacillus thuringiensis* to protect trees from spruce budworm defoliation. *Forest Ecology and Management* 432, 1013–1021.
- García, M., Sturtevant, B.R., Saint-Amant, R., Charney, J.J., Delisle, J., Boulanger, Y., Townsend, P.A., Régnière, J., 2022. Modeling weather-driven long-distance dispersal of spruce budworm moths (*Choristoneura fumiferana*). Part 1: Model description. *Agricultural and Forest Meteorology* 315, 108815.
- Greenbank, D.O., 1963. The analysis of moth survival and dispersal in the unsprayed area. *The Memoirs of the Entomological Society of Canada* 95 (S31), 87–99.
- Greenbank, D.O., Schaefer, G.W., Rainey, R.C., 1980. Spruce Budworm (Lepidoptera: Tortricidae) Moth Flight And Dispersal: New Understanding From Canopy Observations, Radar, and Aircraft. *Memoirs of the Entomological Society of Canada* 112 (S110), 1–49.
- Herms, D.A., 2004. Using degree-days and plant phenology to predict pest activity. *IPM (integrated pest management) of midwest landscapes* 58, 49–59.
- Hirsch, B.T., Visser, M.D., Kays, R., Jansen, P.A., 2012. Quantifying seed dispersal kernels from truncated seed-tracking data. *Methods in Ecology and Evolution* 3, 595–602.
- Ims, R.A., Andreassen, H.P., 2005. Density-dependent dispersal and spatial population dynamics. *Proc. R. Soc. B* 272 (1566), 913–918.
- James, P.M.A., Robert, L.-E., Wotton, B.M., Martell, D.L., Fleming, R.A., 2017. Lagged cumulative spruce budworm defoliation affects the risk of fire ignition in Ontario, Canada. *Ecological Applications* 27 (2), 532–544.
- Jobin, L. J., and C. Coulomb. 1988. Le piège à insectes Multi-Pher®.in R. d. Q. Forêts Canada, Sainte-Foy (Québec). editor. *Canadian Forest Service Publications*.
- Johns, R.C., Bowden, J.J., Carleton, D.R., Cooke, B.J., Edwards, S., Emilson, E.J.S., James, P.M.A., Kneeshaw, D., MacLean, D.A., Martel, V., Moise, E.R.D., Mott, G.D., Norfolk, C.J., Owens, E., Pureswaran, D.S., Quiring, D.T., Régnière, J., Richard, B., Stastny, M., 2019. A conceptual framework for the spruce budworm Early Intervention Strategy: Can outbreaks be stopped? *Forests* 10, 1–19.
- Larroque, J., R. Johns, J. Canape, B. Morin, and P. M. A. James. 2020. Spatial genetic structure at the leading edge of a spruce budworm outbreak: The role of dispersal in outbreak spread. *Forest Ecology and Management* 461:117965-117965.
- Larroque, J., Legault, S., Johns, R., Lumley, L., Cusson, M., Renaut, S., Levesque, R.C., James, P.M.A., 2019. Temporal variation in spatial genetic structure during population outbreaks: Distinguishing among different potential drivers of spatial synchrony. *Evolutionary Applications* 12 (10), 1931–1945.
- Ly, S., Charles, C., Degré, A., 2013. Different methods for spatial interpolation of rainfall data for operational hydrology and hydrological modeling at watershed scale: A review. *Biotechnologie, agronomie, société et environnement* 17, 392–406.
- MacLean, D.A., 2016. Impacts of insect outbreaks on tree mortality, productivity, and stand development. *Canadian Entomologist* 148, S138. S159.
- MacLean, D.A., 2019. Protection Strategy against Spruce Budworm. *Protection strategy against spruce budworm. Forests* 10 (12), 1137.
- MacLean, D.A., Amirault, P., Amos-Binks, L., Carleton, D., Hennigar, C., Johns, R., Régnière, J., 2019. Positive results of an early intervention strategy to suppress a spruce budworm outbreak after five years of trials. *Forests* 10 (5), 448.
- Mair, A., Fares, A., 2011. Comparison of rainfall interpolation methods in a mountainous region of a tropical island. *Journal of hydrologic engineering* 16 (4), 371–383.
- Mazzi, D., Dorn, S., 2012. Movement of insect pests in agricultural landscapes. *Annals of Applied Biology* 160 (2), 97–113.
- Ministère des Forêts, d. l. F. e. d. P. 2015. Aires infestées par la tordeuse des bourgeons de l'épinette au Québec en 2015. Gouvernement du Québec, Direction de la protection des forêts Québec, Canada.
- Ministère des Forêts, d. l. F. e. d. P. 2020. Aires infestées par la tordeuse des bourgeons de l'épinette au Québec en 2020. Gouvernement du Québec, Direction de la protection des forêts Québec, Canada.
- Myers, J.H., 1993. Population outbreaks in forest Lepidoptera. *American Scientist* 81, 240–251.
- Nathan, R., Klein, E., Robledo-Arnuncio, J.J., Revilla, E., 2012. In: *Dispersal Ecology and Evolution*. Oxford University Press, pp. 186–210.
- Nealis, V.G., Régnière, J., 2004. Fecundity and recruitment of eggs during outbreaks of the spruce budworm. *Canadian Entomologist* 136 (4), 591–604.
- Osborne, J. L., and H. D. Loxdale. 2014. Chapter 2 Monitoring insect dispersal : methods and approaches.
- Peltonen, M., Liebhold, A.M., Bjørnstad, O.N., Williams, D.W., 2002. Spatial synchrony in forest insect outbreaks: Roles of regional stochasticity and dispersal. *Ecology* 83 (11), 3120–3129.
- Pureswaran, D.S., Johns, R., Heard, S.B., Quiring, D., 2016. Paradigms in Eastern Spruce Budworm (Lepidoptera : Tortricidae) Population Ecology : A Century of Debate. *Environmental Entomology* 45 (6), 1333–1342.
- Régnière, J., J. Delisle, E. Baucé, A. Dupont, P. Therrien, E. Kettela, L. Cadogan, A. Retnakaran, and K. van Frankenhuyen. 2001. Understanding of Spruce Budworm Population Dynamics: Development of Early Intervention Strategies - in *Boreal Odyssey*. Pages 57-68 *Proceedings of the North American Forest Insect Work Conference*.
- Régnière, J., Nealis, V.G., 2019. Density dependence of egg recruitment and moth dispersal in spruce budworms. *Forests* 10, 1–16.
- Régnière, J., St-Amant, R., 2007. Stochastic simulation of daily air temperature and precipitation from monthly normals in North America north of Mexico. *International Journal of Biometeorology* 51 (5), 415–430.
- Régnière, J., You, M., 1991. A simulation model of spruce budworm (Lepidoptera: Tortricidae) feeding on balsam fir and white spruce. *Ecological Modelling* 54 (3-4), 277–297.
- Régnière, J., St-Amant, R., Duval, P., 2012. Predicting insect distributions under climate change from physiological responses: spruce budworm as an example. *Biological Invasions* 14 (8), 1571–1586.
- Rhainds, M., 2015. Wing wear and body size measurements of adult spruce budworms captured at light traps: inference on seasonal patterns related to reproduction. *Applied Entomology and Zoology* 50 (4), 477–485.
- Rhainds, M., Gries, G., Ho, C.T., Chew, P.S., 2002. Dispersal by bagworm larvae, *Metisa plana*: effects of population density, larval sex, and host plant attributes. *Ecological Entomology* 27, 204–212.
- Rhainds, M., Lavigne, D., Rideout, T., Candau, J.-N., 2019. Temporal variation in abundance of male and female spruce budworms at combinatory associations of light traps and pheromone traps. *Entomologia Experimentalis et Applicata* 167 (6), 526–533.
- Rhainds, M., Lavigne, D., Boulanger, Y., DeMerchant, I., Delisle, J., Mott, J., Rideout, T., Labrecque, A., 2022. I know it when I see it: Incidence, timing and intensity of immigration in spruce budworm. *Agricultural and Forest Entomology* 24 (2), 152–166.
- Risser, P.G., 1987. Landscape ecology: state of the art. *Landscape heterogeneity and disturbance*:3–14.
- Royama, T., 1984. Dynamics of the Spruce Budworm *Choristoneura fumiferana*. *Ecological Monographs* 54, 429–462.
- Sanders, C.J., Wallace, D.R., Lucuik, G.S., 1978. Flight activity of female eastern spruce budworm (Lepidoptera: Tortricidae) at constant temperatures in the laboratory. *The Canadian Entomologist* 110 (6), 627–632.
- Stein, A., Draxler, R.R., Rolph, G.D., Stunder, B.J., Cohen, M., Ngan, F., 2015. NOAA's HYSPLIT atmospheric transport and dispersion modeling system. *Bulletin of the American Meteorological Society* 96, 2059–2077.
- Sturtevant, B.R., Achtemeier, G.L., Charney, J.J., Anderson, D.P., Cooke, B.J., Townsend, P.A., 2013. Long-distance dispersal of spruce budworm (*Choristoneura fumiferana* Clemens) in Minnesota (USA) and Ontario (Canada) via the atmospheric pathway. *Agricultural and Forest Meteorology* 168, 186–200.
- Sturtevant, B.R., Cooke, B.J., Kneeshaw, D.D., MacLean, D.A., 2015. In: *Simulation Modeling of Forest Landscape Disturbances*. Springer International Publishing, Cham, pp. 93–134.
- Tischendorf, L., Fahrig, L., 2000. On the usage and measurement of landscape connectivity. *Oikos* 90 (1), 7–19.
- Volney, W.J.A., Fleming, R.A., 2000. Climate change and impacts of boreal forest insects. *Agriculture, Ecosystems and Environment* 82 (1-3), 283–294.